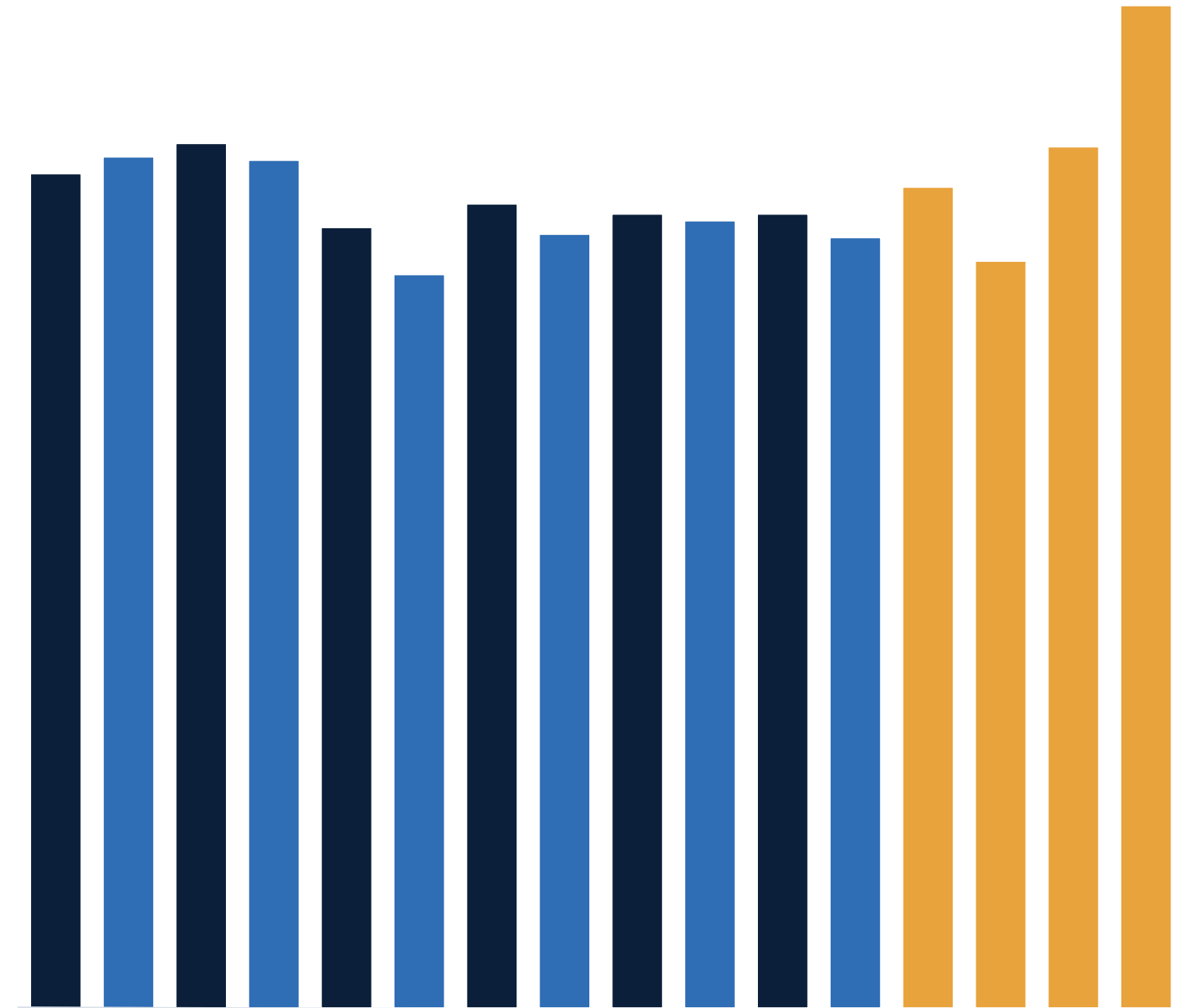


Customer Shopping Behaviour Analysis

Understanding customer spending behaviour and subscription patterns through data



BUSINESS INTELLIGENCE · DATA ANALYTICS · CUSTOMER INSIGHTS

Tools Python · Scikit-learn · XGBoost · Tableau

What can customer behaviour tell us?

01

Behaviour

What patterns characterize customer purchasing behaviour?

02

Segmentation

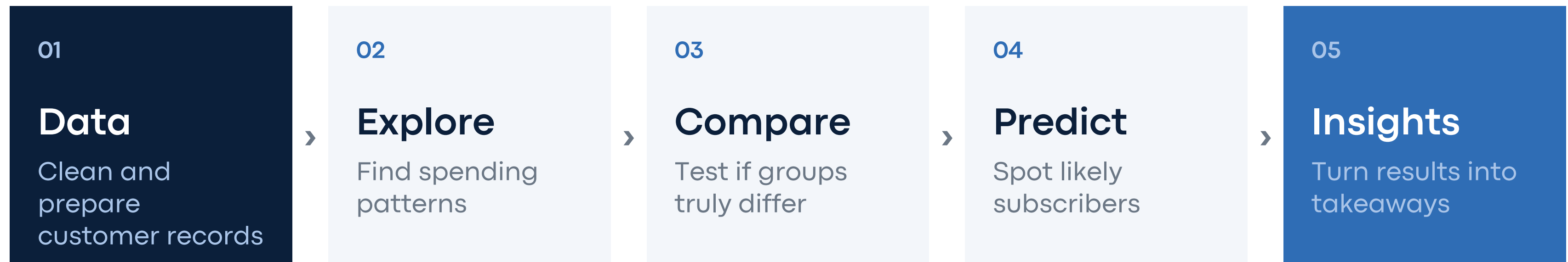
How do customer spending groups differ?

03

Prediction

Can customer behaviour help predict subscription status?

From raw customer data to business insights

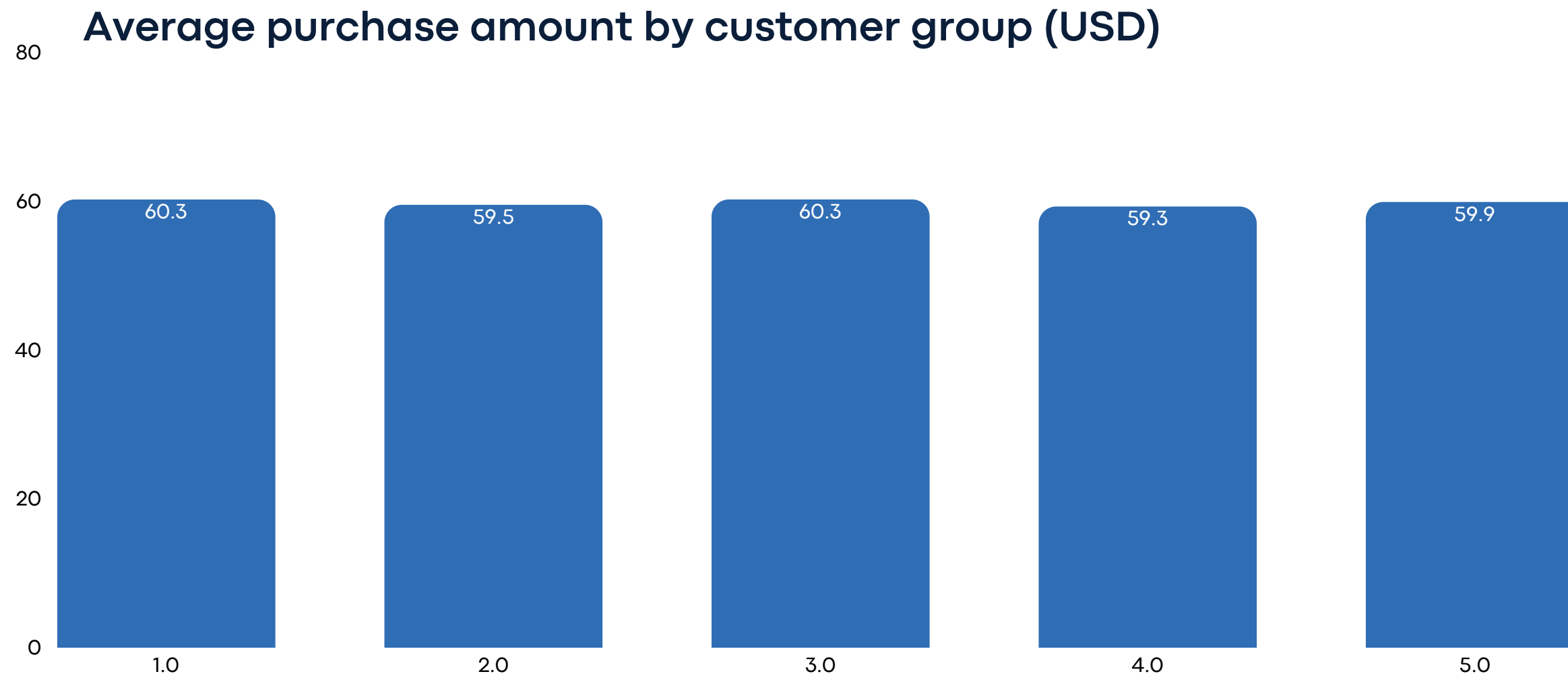


~3,900
customer records

3
business questions

**Python +
Tableau**
tools used

All customer groups spend about \$60 per purchase

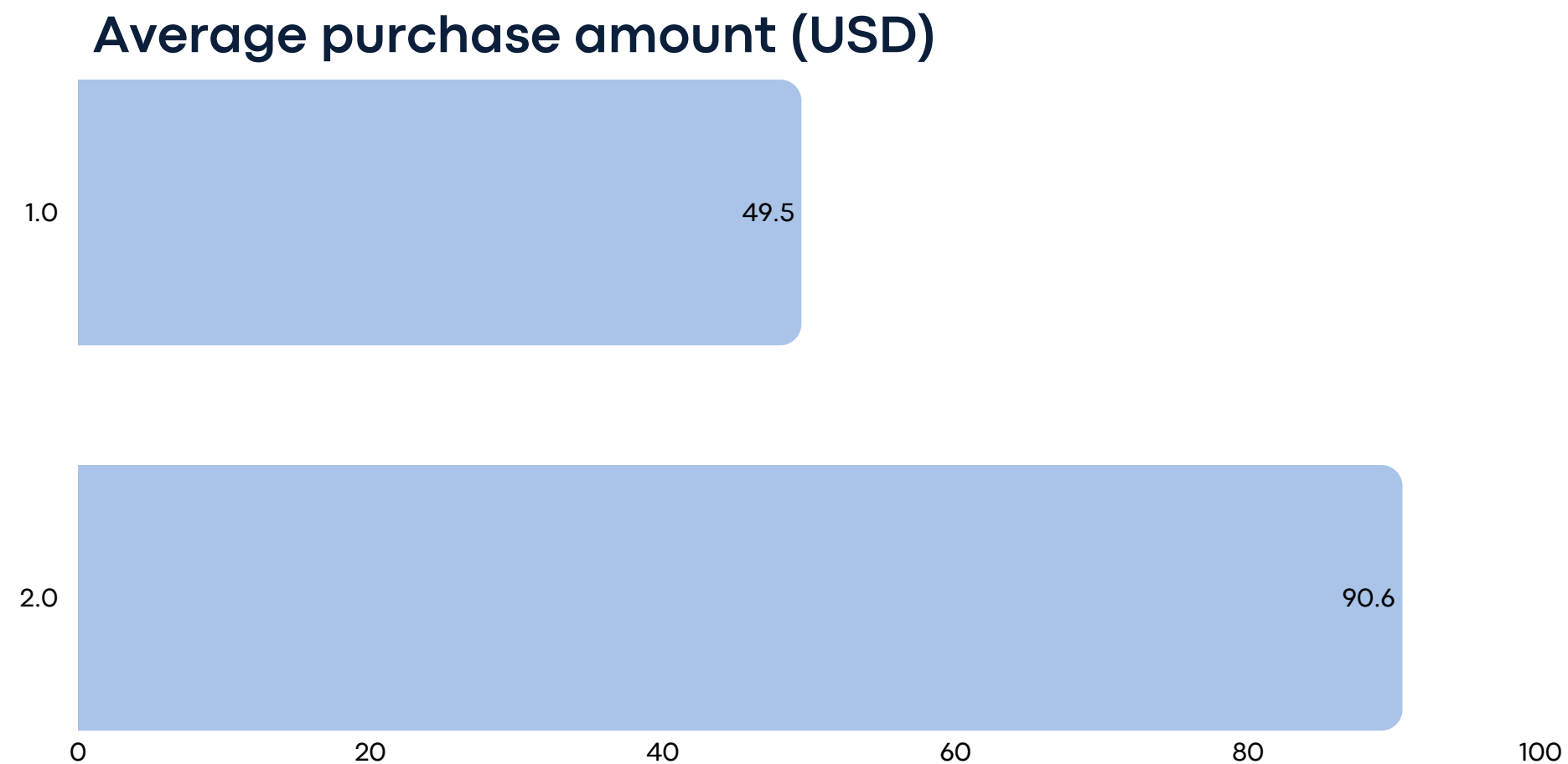


Past purchases vs. spend

0.01
correlation: effectively no link between how often someone bought before and how much they spend now.

Gender, age and purchase history do not explain how much a customer spends.

1 in 4 customers spends far more per purchase



Regular Spender

2,922 customers

\$49.46

High Spender

978 customers

\$90.55

High Spender = top 25% of purchases (above the 75th percentile, \$81). Gap confirmed statistically: $p < 0.0001$ (Mann-Whitney U), Cohen's $d = 3.12$.

High Spenders show substantially higher average purchase values than Regular Spenders.

We can identify subscribers with 86% accuracy

86.41%

Accuracy

of customers classified correctly

100%

Subscriber recall

every actual subscriber was found

66.56%

Subscriber precision

of customers flagged, about 2 in 3 are subscribers

Four prediction methods were compared on an 80/20 train-test split. Logistic Regression performed best.

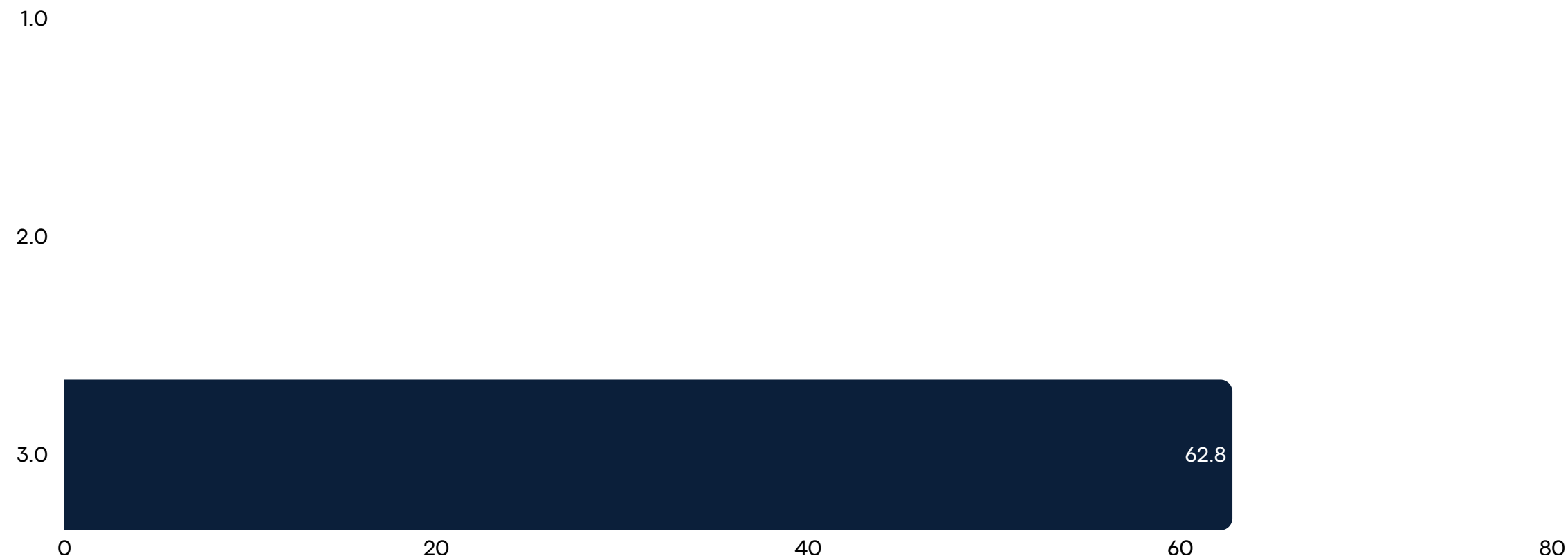
Accuracy alone can mislead, so precision and recall are shown alongside it.

Technical detail: AUC-ROC 91.40% · F1-Macro 84.83% · CV F1-Macro 81.87% · Generalization gap 2.98%

WHAT THE MODEL FOUND

All subscribers are men who used a discount

Share of customers who subscribe (in this dataset)



Do subscribers spend more?

\$59.49

average purchase, subscribers

\$59.87

average purchase, non-subscribers

Subscription is linked to who the customer is and discount use, not to how much they spend.

Three questions, three answers

WHAT WE FOUND

WHAT IT MEANS

01

Behaviour

Customers spend about \$60 per purchase regardless of gender, age or purchase history.

Who a customer is does not tell us how much they will spend.

02

Segmentation

978 High Spenders average \$90.55 vs \$49.46 for Regular Spenders, a statistically significant gap.

High Spenders look like everyone else: 67.4% male vs 68.2%, average age 44.2 vs 44.0.

03

Prediction

Subscribers can be identified with 86.41% accuracy and 100% recall.

All subscribers are men who used a discount, and they spend the same as non-subscribers (\$59.49 vs \$59.87).

What we recommend doing next

WHAT TO DO

WHY (EVIDENCE)

01

Look beyond demographics

Don't target High Spenders by gender, age or purchase history alone. Study what else sets the top 25% apart.

Average spend is about \$60 in every gender and age group. High Spenders are 67.4% male vs 68.2% for others.

02

Open subscriptions to women

Find out why female customers do not subscribe, and pilot a subscription offer for them.

0 of 1,248 female customers subscribe, vs 39.7% of male customers (1,053 of 2,652).

03

Test discounts before scaling

Run a controlled test to see whether discounts drive subscriptions before expanding them.

All 1,053 subscribers used a discount. 62.8% of men who used one subscribe, vs 0% of men who did not.

These patterns show association in this dataset, not cause. Confirm how discounts and subscriptions are linked before acting.